

BENJAMIN MILLER

(925) 270-9677 | bem002@ucsd.edu | CA - Bay Area
LinkedIn | GitHub | Portfolio Website

EDUCATION

Computer Science B.S. - University of California, San Diego (UCSD) Sept. 2023 – June 2027
GPA: 3.8/4.0, Jacobs School of Engineering, UC San Diego Scholar's Society
Regent Scholar (Merit-based, top 1.5% of class)

Relevant Coursework: Advanced Data Structures and Algorithms (**C++**), Operating Systems, Embedded Programming (**C**, **ARM Assembly**, **Microcontrollers**), Software Engineering, Components and Design Techniques for Digital Systems (**SystemVerilog**, **Register Transfer Logic - RTL**), Object Oriented Design

SKILLS

- Programming Languages: **C**, **C++**, **Python**, **ARM Assembly**, **Java**, **SystemVerilog**
- Embedded Systems: **FreeRTOS**, **I2C**, **UART**, **PWM**, **GPIO**, **ESP32**, **Servos**, **Raspberry Pi**
- Machine Learning: **NumPy**, **PyTorch**, **Pandas**, **scikit-learn**, **Computer Vision**
- Tools & DevOps: **Linux**, **Git Version Control**, **Docker**, **Bash**, **CI/CD**, **Agile/Scrum**, **Makefile**

EXPERIENCE

Embedded Software Engineer Intern Sept. 2025 - Dec. 2025, Jun. 2026 - Present
Western Digital Irvine, CA

- Developed and debugged **embedded C++ firmware** within a **multi-threaded** codebase, implementing and validating system behaviors through unit and integration testing.
- Investigated firmware bugs involving asynchronous power offs, error handling, and system state transitions, identifying root causes through debugging tools and code-level analysis.
- Contributed to the development of firmware for next generation HAMR (Heat-Assisted Magnetic Recording) storage devices to improve device reliability and performance.
- Strengthened embedded C++ error-handling paths in a production firmware codebase, improving clarity and reliability of system error states.
- Leveraged **Agentic AI** tools and **MCP (model context protocol)** integrations to automate development workflows resulting in higher efficiency.
- Developed an internal tool to accelerate remote, server-based firmware integration testing using a **Python** microservice and full-stack application, validated through successful end-to-end demonstrations.

Software Developer Intern Apr. 2025 - Aug. 2025
Center for Applied Internet Data Analysis San Diego, CA

- Created useful automations using **Python** in a **Linux** environment resulting in a 70% decrease on average in manual labor required for workflows, such as content updates across the center's project sites.
- Enhanced maintainability and usability using **JavaScript** and **Python**, as demonstrated by successful contributions to the websites' codebase using **Git** for version control in large-scale site environments.

Resident Advisor at COSMOS Jul. 2024 - Aug. 2024
Jacobs School of Engineering La Jolla, CA

- Provided mentorship for the Video Game Programming and AI Design group, as demonstrated by meaningful interactions and personalized advice to empower future engineers.

LEADERSHIP

Electronic Props Designer Mar. 2026 - Present
Star Wars Club at UCSD La Jolla, CA

- Designed **embedded systems** and **C++ firmware** for props for film making and on-campus events using a real time operating system (**RTOS**) and microcontrollers resulting in screen accurate prop recreations.

Software Engineering Lead & Vice President Jun. 2024 - Jun. 2026
Themed Entertainment Association at UCSD La Jolla, CA

- Led development of **embedded systems** for themed attraction software on **microcontrollers** and single board computers, running themed installations that were attended by 100+ guests campuswide.

PROJECTS

Taro The Talking Bird - Interactive Robotic Figure Dec. 2025
C/C++, I2C, Advanced Linux Sound Architecture, Multi-Threading, Raspberry Pi Demo Video | Project Page | Github

- Improved embedded system performance by designing a multithreaded architecture for motor actuation and concurrent audio processing, ensuring less than 5ms motion latency.
- Built a **C++** real-time audio pipeline, synchronizing live voice actor speech input with the robotic mouth over **I2C**, demonstrated in **16K+ social media views** and 100+ live demos.